

MATTEO SAPONATI

Research Scientist - Project Lead

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🏠 Zürich, Switzerland



Experience

Research Scientist

2026 - present

📍 Tufa Labs, Zurich (CH)

🏢 Full-time, on-site

- Designed pretraining and synthetic data pipelines to build very small, highly capable reasoning models.
- Developed and tested automatic ML research workflows with coding agents.
- Led research projects on self-distillation post-training algorithms and supervised interns.

Consultant - Neuromorphic Computing

2025 - 2025

📍 Neuronova, Milan (IT)

🏢 Part-time, remote

- Conducted an R&D project on Edge AI for audio applications.
- Provided expert consulting on neuromorphic training algorithms and hardware–software codesign principles
- Worked independently as an external consultant, coordinating with the Neuronova team throughout the project lifecycle.

Postdoctoral Research Scientist

2023 - 2025

📍 Institute of Neuroinformatics, ETH/UZH, Zurich (CH)

🏢 Full-time, on-site

- Secured competitive research funding to design advanced training algorithms for edge AI.
- Led research and deployment of Large Language Models, Spiking Neural Networks, and Recurrent Neural Networks.
- Published 4 peer-reviewed papers; presented at 8 international conferences.
- Supervised M.Sc. and Ph.D. students from ETH Zurich, University of Zurich, and ZHAW Center for Artificial Intelligence.

PhD researcher

2020 - 2023

📍 Max-Planck Institute for Brain Research and Ernst Strüngmann Institute for Neuroscience, Frankfurt Am Main (DE)

🏢 Full-time, on-site

- Designed learning algorithms for Spiking Neural Networks applied to Machine Learning and Neuroscience.
- Published 3 peer-reviewed articles; presented at 6 international conferences.
- Led data analysis projects; Coordinated scientific seminars.

Assistant Research Scientist

2019

📍 Institute des Neurosciences des Systèmes, Aix-Marseille University, Marseille (FR)

🏢 Full-time, on-site

- Conducted research in Computational Neuroscience.
- Developed models to describe neuronal coupling across spatial scales.
- Investigated theoretical limits using The Virtual Brain platform (TVB).

Research Intern

2018

📍 Barcelona Biomedical Research Park, Barcelona (ESP)

🏢 Full-time, on-site

- Published 2 papers and presented research at international conferences.

Education

- 2020 - 2023 **Ph.D. in Neuroinformatics**
Highest Honors (top 5%) - Donders Centre for Neuroscience, Radboud University (NL)
- 2016 - 2018 **M.Sc. in Physics**
110/110 - Department of Physics, University of Pisa (IT)
- 2011 - 2016 **B.Sc. in Physics**
94/110 - Department of Physics, University of Pisa (IT)

Skills

- Coding Skills** Python, PyTorch, Codex/Claude Code, Linux, LaTeX, C++
- Research Skills** Machine Learning, LLM training and inference, Mechanistic Interpretability, Mathematical Modelling, Critical Thinking, Public Speaking, Teamwork, Problem Solving
- Language Skills** Italian (Mother tongue), English (Business Fluent), Portuguese (Business Intermediate), French (Basic)

Research

- Haag, J., Metzner, C., Zendrikov, D., Indiveri, G., Grewe, B., De Luca*, C., & **Saponati***, M. (2026). Mixed-signal implementation of feedback-control optimizer for single-layer spiking neural networks. *IEEE International Symposium on Circuits and Systems*. <https://doi.org/10.48550/arXiv.2603.24113>
- Saponati, M.**, & Vinck, M. (2026). Inhibitory feedback enables predictive learning of multiple sequences in neural networks. *Neural Computation*, 38(4), 471–498. <https://doi.org/https://doi.org/10.1162/NECO.a.1504>
- Stalder, N., Grewe, B., **Saponati***, M., & Aceituno Vilimelis*, P. (2026). A combination of noise and bilateral filters achieve supralinear and scalable adversarial robustness in cnns. *The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2026*. Retrieved June 3, 2026, from <https://cvpr.thecvf.com/virtual/2026/poster/37609>
- Saponati, M.**, De Luca, C., Indiveri, G., & Grewe, B. (2025). A feedback control optimizer for online and hardware-aware training of spiking neural networks. *2025 Neuro-Inspired Computational Elements Conference (NICE)*. <https://doi.org/10.1109/NICE65350.2025.11065428>
- Saponati***, M., Sager*, P., Aceituno Vilimelis, P., Stadelmann, T., & Grewe, B. (2025). The underlying structures of self-attention: Symmetry, directionality, and emergent dynamics in transformer training. *Proceedings of the 41st International Conference on Machine Learning*. <https://doi.org/10.48550/arXiv.2502.10927>
- Saponati, M.**, & Vinck, M. (2023). Sequence anticipation and spike-timing-dependent plasticity emerge from a predictive learning rule. *Nature Communications*, 14(1), 4985. <https://doi.org/10.1038/s41467-023-40651-w>
- Saponati, M.**, Garcia-Ojalvo, J., Cataldo, E., & Mazzoni, A. (2022). Thalamocortical Spectral Transmission Relies on Balanced Input Strengths. *Brain Topography*, 35(1), 4–18. <https://doi.org/10.1007/s10548-021-00851-3>
- Spyropoulos, G., **Saponati, M.**, Dowdall, J. R., Schölvinck, M. L., Bosman, C. A., Lima, B., Peter, A., Onorato, I., Klon-Lipok, J., Roese, R., Neuenschwander, S., Fries, P., & Vinck, M. (2022). Spontaneous variability in gamma dynamics described by a damped harmonic oscillator driven by noise. *Nature Communications*, 13(1), 2019. <https://doi.org/10.1038/s41467-022-29674-x>
- Saponati, M.**, Garcia-Ojalvo, J., Cataldo, E., & Mazzoni, A. (2019). Integrate-and-fire network model of activity propagation from thalamus to cortex. *Biosystems*, 183, 103978. <https://doi.org/10.1016/j.biosystems.2019.103978>

Grants and Awards

- Jan 2024 - Jan 2026 **ETH Postdoctoral Fellowship**
ETH Zurich Postdoctoral Fellowship programme (Zürich, CH)
- Mar 2023 **Cosyne Presenters Travel Grant**
Cosyne Conference 2023 (Montreal, CA)
- Sep 2019 - Sep 2023 **IMPRS Research Fellowship**
International Max Planck Research School (IMPRS) for Neural Circuits, MPI for Brain Research, Frankfurt am Main (DE)
- Jul 2018 - Aug 2018 **Erasmus+ Grant**
Erasmus program (EU)